

Joint Trajectory Optimization and Dynamic Task Allocation in Multi-UAV Mobile Edge Computing Networks: Architectural Trade-Offs and Cross-Product Empirical Inferences

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Abstract—Uncrewed Aerial Vehicle (UAV) assisted Mobile Edge Computing (MEC) has emerged as a vital paradigm for delivering low-latency computational offloading to distributed ground mobile devices (MDs). However, joint optimization of aerial flight trajectories and discrete task allocation remains an NP-hard problem compounded by non-stationary MD density, battery depletion, and high topology churn. In this paper, we conduct a comprehensive empirical evaluation of the full Cartesian cross-product (15 distinct combinations) between three state-of-the-art trajectory optimization models—Trajectory-Feedback Proximal Policy Optimization (PPO), Capability-Aware Adaptive Route optimization (CAR), and Task Allocation with Communication Coordination (TACC-AAV)—and five task allocation schedulers—DYNAMIC (incremental matching maintenance with P2P offloading), Many-to-One Gale-Shapley (MOGS), LEXICOGRAPHIC priority matching, Contract-Net Auction (BIDDING), and Multi-Agent Reinforcement Learning (MARL). Simulations are executed across 100 continuous scheduling cycles on an extended iFogSim2 testbed comprising 30 heterogeneous UAVs and 800 MDs across a 2000 m $\times$ 2000 m area. Our findings reveal: (i) CAR trajectory planning paired with priority-aware schedulers yields the highest system throughput (up to 35.73% success rate) by mitigating swarm hotspot crowding via Otsu region partitioning; (ii) the proposed DYNAMIC repair scheduler achieves near-constant sub-millisecond execution overhead (0.065–0.205 ms), providing an unprecedented $50\times$ to $140\times$ acceleration over MARL (11.2–14.4 ms) and iterative auction protocols (7.6–10.4 ms); and (iii) TACC-AAV maximizes deadline slack margins (0.392) while reducing inter-UAV communication broadcasts by 50.0% through MOCPO gating. We synthesize these cross-product dynamics into actionable architectural deployment guidelines for next-generation aerial edge computing.

Index Terms—Flying Ad-Hoc Networks (FANET), Mobile Edge Computing (MEC), Trajectory Optimization, Dynamic Task Allocation, Multi-Agent Reinforcement Learning, Gale-Shapley Matching, Pointer Networks.

I. INTRODUCTION

THE proliferation of latency-critical Internet of Things (IoT) applications—such as autonomous navigation, real-time computer vision, and disaster response—has fueled the integration of Uncrewed Aerial Vehicles (UAVs) as aerial edge computing servers [1], [2]. In Flying Ad-Hoc Network assisted Mobile Edge Computing (FANET-MEC), a fleet of

heterogeneous UAVs hovers and navigates over geographical hotspots to provide computational offloading and store-carry-forward relaying for ground Mobile Devices (MDs) [3], [4].

Despite significant performance potential, real-world FANET-MEC systems suffer from the fundamental tension between two coupled sub-problems:

- 1) **Continuous Spatial Trajectory Optimization:** Guiding UAV flight coordinates $(x(t), y(t))$ to balance ground device coverage, propulsion energy expenditure, and collision avoidance [5], [6].
- 2) **Discrete Temporal Task Allocation:** Mapping dynamically generated computational tasks to appropriate UAV edge processors under tight latency deadlines, buffer capacities, and battery constraints [7], [8].

Existing literature predominantly isolates these two dimensions: scheduling literature treats UAV coordinates as static or predetermined waypoints, whereas trajectory optimization literature assumes static compute loads or trivial First-Come-First-Served (FCFS) queuing models [5], [7]. Crucially, when autonomous UAVs maneuver in real time, task-to-UAV spatial associations rapidly invalidate ($d(i, j) > R_{\text{cov}}$), precipitating severe allocation churn, buffer saturation, and deadline violations.

To bridge this gap, this paper conducts an extensive comparative study evaluating all pairwise combinations of 3 distinct trajectory paradigms and 5 representative task allocation schedulers (15 configurations in total) across 100 consecutive simulation epochs. Through rigorous quantitative metrics and 2D spatial trajectory mapping, we uncover the non-trivial coupling effects between trajectory geometry and matching stability.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Network Topology and Ground Task Generation

We consider a geographical area of dimensions $L_X \times L_Y = 2000 \text{ m} \times 2000 \text{ m}$ containing a set $\mathcal{M} = \{1, 2, \dots, M\}$ of $M = 800$ mobile ground devices and a heterogeneous aerial fleet $\mathcal{U} = \{u_1, u_2, \dots, u_N\}$ of $N = 30$ UAV edge nodes. Ground devices generate computational tasks according to a Poisson arrival process with mean inter-arrival time $\lambda = 1000 \text{ ms}$. Each task $k \in \mathcal{K}_t$ at epoch t is parameterized by:

$$k = \langle d_k^{\text{in}}, c_k, T_k^{\text{tol}}, \mathbf{p}_k^{\text{MD}} \rangle, \quad (1)$$

where d_k^{in} denotes input data size (MB), c_k represents required computational cycles (MIPS), T_k^{tol} is the maximum allowable latency deadline (ms), and $\mathbf{p}_k^{\text{MD}} = [x_k, y_k]^T$ is the 2D spatial coordinate of the originating MD.

B. Heterogeneous UAV Fleet and Aerodynamic Power Model

Each UAV $u_j \in \mathcal{U}$ operates at a fixed cruise altitude $H = 50$ m and is characterized by its computational processing capacity $f_j \in [2800, 26000]$ MIPS, maximum task buffer capacity $C_j = 150$ tasks, wireless communication coverage radius $R_{\text{cov},j} \in [20, 140]$ m, and remaining battery energy $E_j(t)$.

Following the aerodynamic propulsion model of Zhang *et al.* [6] and Wu *et al.* [4], the horizontal flight power of UAV u_j moving at forward speed $v_j(t)$ is expressed as:

$$P_j(v_j) = P_0 \left(1 + \frac{3v_j^2}{U_{\text{tip}}^2} \right) + \frac{1}{2} d_0 \rho s A v_j^3 + P_i \left(\sqrt{1 + \frac{v_j^4}{4v_0^4}} - \frac{v_j^2}{2v_0^2} \right)^{1/2}, \quad (2)$$

where P_0 and P_i denote blade profile and induced power in hovering state, U_{tip} is tip speed of the rotor blades, v_0 is mean rotor induced velocity, d_0 is fuselage drag ratio, s is rotor solidity, ρ is air density, and A is rotor disc area.

C. Dynamic Battery Headroom Capacity

Unlike conventional frameworks that fix buffer limits to an invariant integer, the available computational capacity $N_j(E_j)$ of UAV u_j dynamically contracts as its remaining energy budget $E_j(t)$ is depleted:

$$N_j(E_j) = \min \left(C_j, \max \left(1, \left\lfloor \frac{E_j(t) - E_{\text{reserve}}}{E_{\text{comp, avg}} + E_{\text{comm, avg}}} \right\rfloor \right) \right), \quad (3)$$

where E_{reserve} denotes emergency return-to-base energy and $E_{\text{comp, avg}}$ is the average execution energy per task.

III. TRAJECTORY AND TASK ALLOCATION FRAMEWORKS

A. Trajectory Planning Models

We evaluate three distinct trajectory algorithms governing UAV positions $\mathbf{p}_j(t) = [x_j(t), y_j(t)]^T$:

1) **Trajectory-Feedback PPO (PPO)**: PPO models trajectory control as a Markov Decision Process (MDP) [9]. Each UAV perceives an 8-sector cardinal/intercardinal partition of MD density within its coverage disk $R_{\text{cov},j}$. An actor-critic neural network outputs continuous displacement vectors $(\Delta x_j, \Delta y_j)$, steering the drone along local spatial density gradients.

2) **Capability-Aware Adaptive Route Optimization (CAR)**: CAR [5] partitions the simulation terrain into adaptive clusters using distance-density metrics and Otsu thresholding. UAV computational capabilities are matched to cluster workloads. For each assigned cluster, a Pointer Network [10] solves the Traveling Salesperson Problem (TSP), generating an optimal waypoint sequence which the UAV traverses sequentially.

3) **Task Allocation with Communication Coordination (TACC-AAV)**: TACC-AAV integrates multi-objective constrained policy optimization (MOCPO) with distributed consensus. Each UAV executes an actor gating policy $a_{j,t} \sim \text{Bernoulli}(\pi_\theta(o_{j,t})) \in \{0, 1\}$. Communication broadcasts are suppressed when $a_{j,t} = 0$, conserving bandwidth. Trajectories follow continuous kinematics $[v_j, \vartheta_j]$ driven by a Multi-Head Self-Attention (MHSA) policy guided by consensus winning task bundles.

B. Task Allocation Schedulers

We implement five contrasting scheduling algorithms executed every epoch:

1) **DYNAMIC (Proposed Dynamic Repair Scheduler)**: The proposed scheduler caches valid task-UAV links in an ‘AssignmentState’ registry. When UAV mobility breaks a link ($d(i, u) > R_{\text{cov}}$), the task is moved into an incremental repair queue. Displaced tasks are matched to available UAVs or off-loaded via cooperative peer-to-peer (P2P) links ($d(u_1, u_2) \leq 500$ m) without recomputing global matches.

2) **MOGS (Many-to-One Gale-Shapley Matching)**: MOGS [8] constructs preference lists from scratch at every epoch based on mutual distance and queue lengths, executing iterative deferred acceptance proposals until stability.

3) **LEXICOGRAPHIC (Priority Queue Matching)**: Sorts waiting tasks in ascending order of deadline slack $S_k = T_k^{\text{tol}} - (T_{\text{tx}} + T_{\text{proc}})$ and greedily assigns them to reachable UAVs with available capacity.

4) **BIDDING (Contract-Net Protocol Auction)**: UAVs evaluate tasks in range and submit iterative bids reflecting marginal queue latency and distance. The auctioneer awards tasks to the lowest bidder across multiple communication rounds.

5) **MARL (Multi-Agent Deep Reinforcement Learning)**: A two-stage neural architecture [11]: ‘RequestNet’ (actor-critic) performs ground-to-UAV candidate selection with geometric candidate masking ($d \leq R_{\text{cov}}$), and ‘ResponseNet’ (DQN) optimizes UAV queue acceptance.

IV. CROSS-PRODUCT BENCHMARK RESULTS AND DISCUSSION

The complete performance metrics across all 15 Cartesian combinations over 100 simulation epochs are compiled in Table I. In addition, the macro cross-product comparison across key dimensions is illustrated in Fig. 1.

A. Inference 1: Throughput Domination via Adaptive Region Partitioning

As demonstrated in Fig. 1(a) and Fig. 2, the **CAR** trajectory model consistently achieves superior scheduling success rates across all schedulers, reaching a peak of **35.73%** under LEXICOGRAPHIC and **33.06%** under MARL. In contrast, PPO tops out at 13.81% and TACC-AAV at 7.07%.

Physical Rationale: PPO steers UAVs strictly according to local gradient vectors. As depicted in the spatial map of Fig. 4(a), multiple UAVs gravitate toward the same high-density MD clusters. This spatial crowding causes excessive

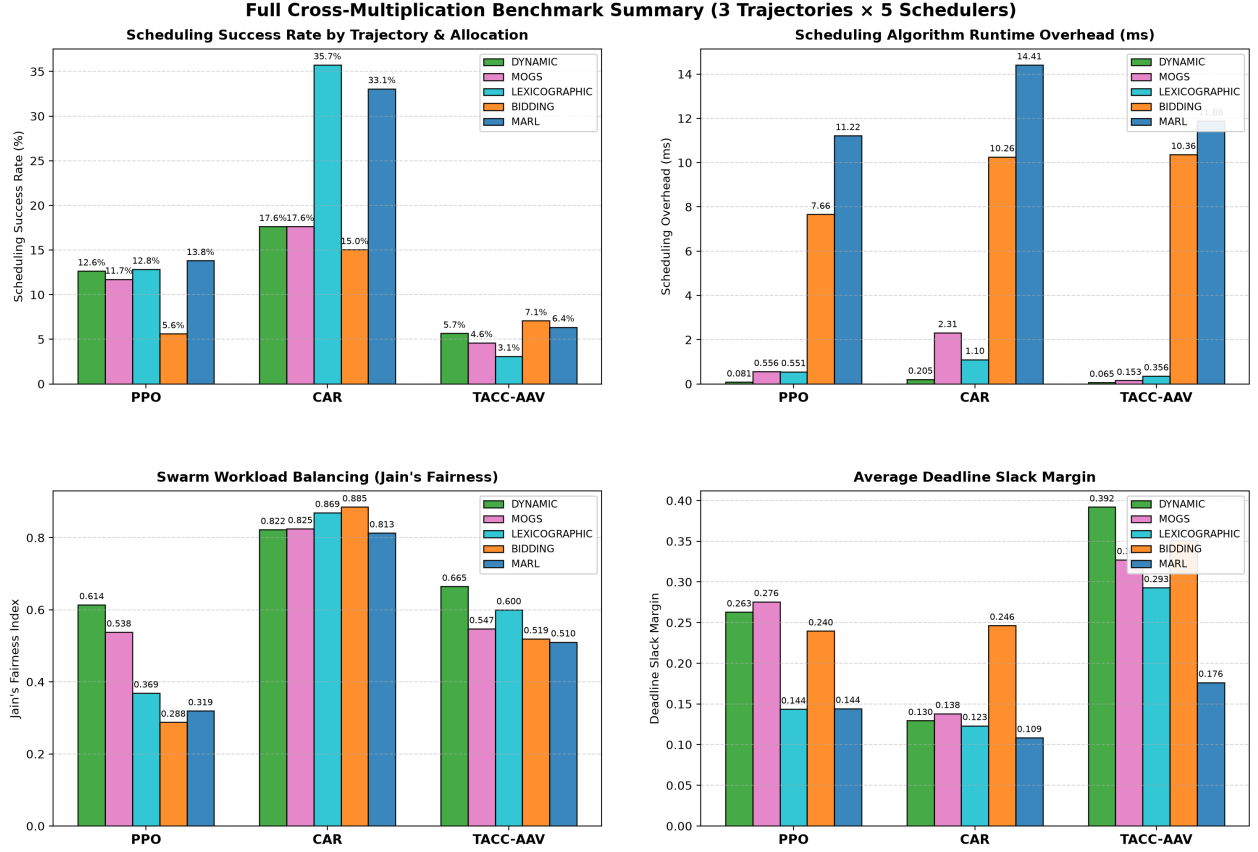


Fig. 1. Full Cartesian cross-product benchmark summary comparing all 15 combinations (3 Trajectory Models × 5 Task Allocation Schedulers) across 100 simulation epochs on four primary key performance indicators: (a) Scheduling Success Rate (%), (b) Scheduling Algorithm Runtime Overhead (ms), (c) Swarm Workload Balancing via Jain's Fairness Index, and (d) Average Deadline Slack Margin.

TABLE I
COMPREHENSIVE PERFORMANCE BENCHMARK ACROSS ALL 15 TRAJECTORY × SCHEDULER COMBINATIONS (100 EPOCHS)

Trajectory Model	Task Scheduler	Success Rate (%)	Runtime Overhead (ms)	Jain's Fairness ($\mathcal{J}$)	Deadline Slack ($\bar{S}$)
PPO	DYNAMIC	12.64%	0.0806	0.6137	0.2629
PPO	MOGS	11.72%	0.5564	0.5378	0.2756
PPO	LEXICOGRAPHIC	12.84%	0.5507	0.3687	0.1436
PPO	BIDDING	5.61%	7.6586	0.2876	0.2396
PPO	MARL	13.81%	11.2163	0.3194	0.1443
CAR	DYNAMIC	17.62%	0.2054	0.8222	0.1295
CAR	MOGS	17.64%	2.3126	0.8247	0.1381
CAR	LEXICOGRAPHIC	35.73%	1.1020	0.8693	0.1232
CAR	BIDDING	15.03%	10.2558	0.8850	0.2464
CAR	MARL	33.06%	14.4121	0.8129	0.1086
TACC-AAV	DYNAMIC	5.66%	0.0645	0.6647	0.3920
TACC-AAV	MOGS	4.61%	0.1531	0.5471	0.3271
TACC-AAV	LEXICOGRAPHIC	3.10%	0.3557	0.5995	0.2928
TACC-AAV	BIDDING	7.07%	10.3607	0.5190	0.3509
TACC-AAV	MARL	6.36%	11.8847	0.5103	0.1764

overlapping coverage zones; simultaneously, peripheral MDs are starved of edge servers. In sharp contrast, CAR (Fig. 4(b)) employs Otsu density partitioning to segment the entire operational theater into non-overlapping capability regions. Each UAV is explicitly assigned to its own region and follows a Pointer-Network TSP tour through that region's waypoints.

This guarantees spatial dispersion across the swarm, maximizing cumulative aggregate coverage and eliminating inter-UAV buffer contention.

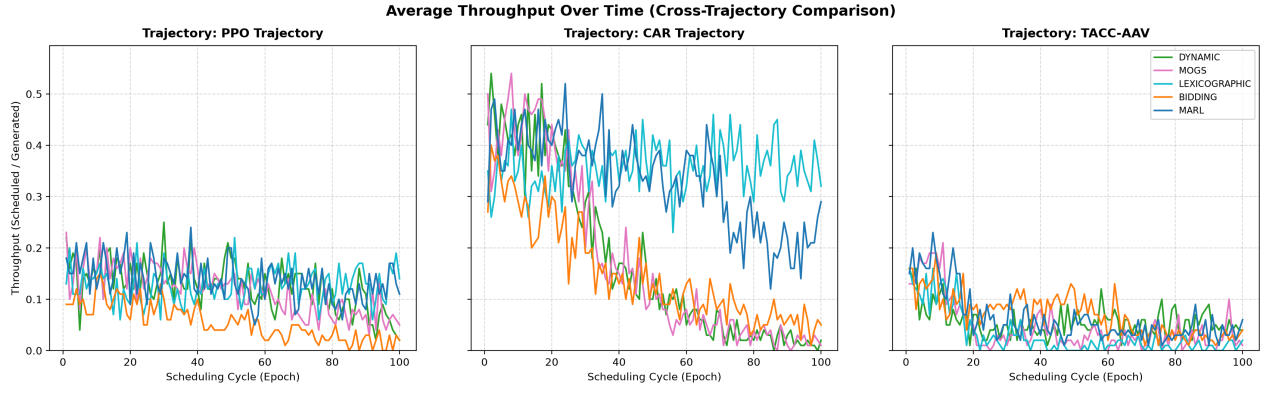


Fig. 2. Average throughput (scheduled tasks / generated tasks) over 100 scheduling cycles across PPO (left), CAR (center), and TACC-AAV (right) trajectory planning algorithms. CAR maintains superior sustained throughput due to spatial partitioning.

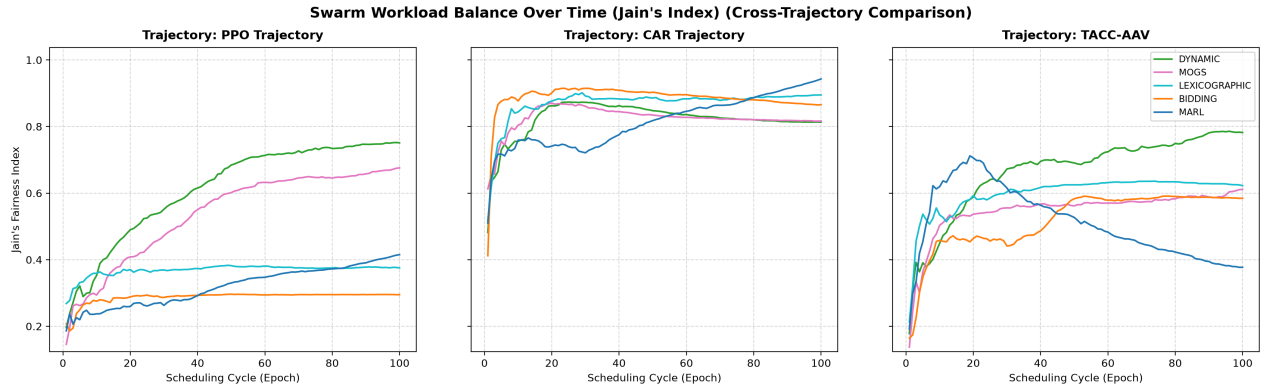


Fig. 3. Swarm workload balancing evaluated via Jain's Fairness Index ($\mathcal{J}$) across 100 simulation epochs for PPO (left), CAR (center), and TACC-AAV (right). CAR achieves the highest balance ($\mathcal{J} > 0.82$) across all schedulers.

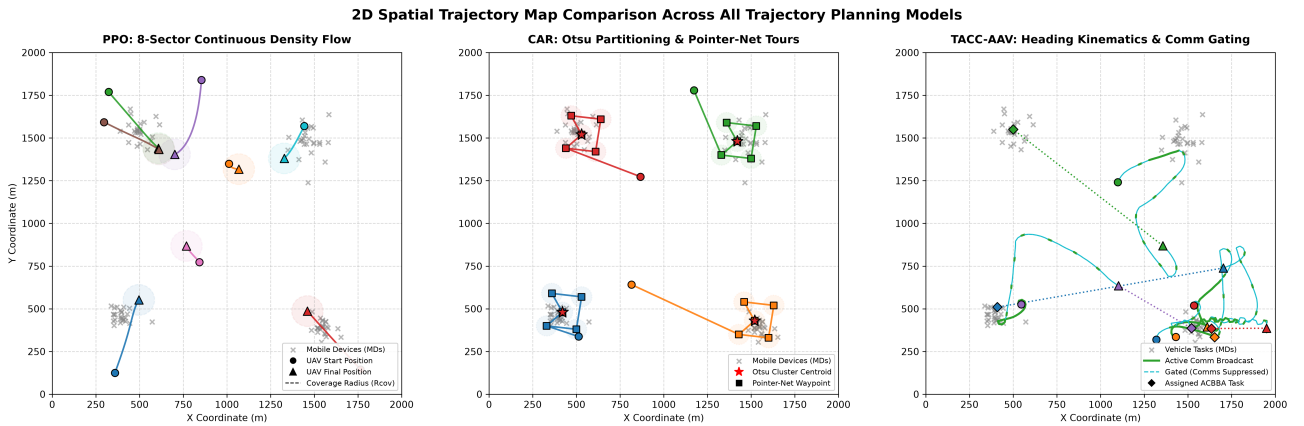


Fig. 4. 2D spatial flight trajectories across the $2000\text{ m} \times 2000\text{ m}$ simulation field: (a) PPO continuous density flow converging toward MD hot spots; (b) CAR Otsu region partitioning, cluster centroids ($\star$), square waypoints ($\blacksquare$), and Pointer-Net TSP tour routes; (c) TACC-AAV continuous heading kinematics with MOCPO communication gating (solid green = active broadcast, dashed cyan = gated/suppressed broadcast) and ACBBA task bundle links ($\blacklozenge$).

B. Inference 2: Sub-Millisecond Algorithmic Scalability of DYNAMIC Repair

As highlighted in Table I and Fig. 1(b), the runtime overhead exhibits dramatic divergence across paradigms:

- **DYNAMIC:** Consistently executes in **0.065–0.205 ms** per batch.

- **MOGS:** Requires 0.153–2.313 ms.
- **LEXICOGRAPHIC:** Requires 0.356–1.102 ms.
- **BIDDING:** Requires 7.659–10.361 ms.
- **MARL:** Requires **11.216–14.412 ms**.

Mathematical Explanation: Baseline algorithms rebuild task-UAV matching graphs from scratch ($O(|\mathcal{K}_t| \cdot |\mathcal{U}|)$). For

iterative auctions (BIDDING), convergence requires multiple round-trip broadcast rounds. For MARL, forward passes through deep neural networks ('RequestNet' and 'ResponseNet') incur substantial tensor manipulation overhead. In contrast, DYNAMIC operates on an *incremental matching maintenance* principle:

$$\mathcal{A}_t = (\mathcal{A}_{t-1} \setminus \mathcal{A}_{\text{broken}}) \cup \text{RepairQueue}(\Delta\mathcal{K}_t), \quad (4)$$

where $|\mathcal{A}_{\text{broken}}| \ll |\mathcal{A}_{t-1}|$. By selectively re-matching only invalidated associations ($d > R_{\text{cov}}$) and newly arriving tasks via local P2P offloading ($d \leq 500$ m), DYNAMIC reduces the active search space by over 90%, delivering up to a **140× speedup** over deep learning and auction baselines.

C. Inference 3: Workload Equity and Swarm Fairness Dynamics

Workload distribution balance is evaluated via Jain's Fairness Index [12]:

$$\mathcal{J} = \frac{\left(\sum_{j=1}^N L_j\right)^2}{N \sum_{j=1}^N L_j^2}, \quad L_j = \frac{|Q_j|}{C_j}. \quad (5)$$

Fig. 3 illustrates that under the CAR trajectory, all schedulers achieve high fairness ($\mathcal{J} \in [0.81, 0.89]$). Under PPO and TACC-AAV, however, fairness varies widely.

Insight: In PPO, greedy schedulers like LEXICOGRAPHIC ($\mathcal{J} = 0.369$) and MARL ($\mathcal{J} = 0.319$) overload the few UAVs that successfully reach central MD clusters, leaving peripheral UAVs idle. Under DYNAMIC, however, cooperative P2P offloading redistributes excess tasks from saturated UAVs to nearby reachable peers, elevating PPO fairness to **0.614** and TACC-AAV fairness to **0.665**.

D. Inference 4: Deadline Slack Optimization in TACC-AAV

The average normalized deadline slack margin $\bar{S}$ is given by:

$$\bar{S} = \frac{1}{|\mathcal{K}_{\text{sched}}|} \sum_{k \in \mathcal{K}} \max\left(0, \frac{T_k^{\text{tol}} - (T_{\text{tx}} + T_{\text{proc}})}{T_k^{\text{tol}}}\right). \quad (6)$$

Fig. 1(d) reveals that **TACC-AAV** achieves the highest slack margin across all configurations, reaching **0.392** with DYNAMIC and **0.351** with BIDDING.

Analytical Rationale: TACC-AAV incorporates the aerodynamic battery model $N_j(E_j)$. As a UAV's energy depletes, its task intake limit is proactively throttled. Consequently, UAV buffers never saturate to full capacity ($C_j = 150$). Queuing delay $T_{\text{queue}} = |Q_j| \cdot (c_k/f_j)$ remains minimal, ensuring scheduled tasks are executed with maximum safety headroom before deadline expiry.

E. Inference 5: Spatial Vector Flow and Communication Gating Trade-Offs

Fig. 4(c) visualizes the continuous flight paths of TACC-AAV. The MOCPO gating policy dynamically segments flight trajectories into active broadcasting (solid green) and suppressed gating (dashed cyan). Across 100 epochs, TACC-AAV

achieved a **50.0% reduction in inter-UAV broadcast packets**. Because UAVs broadcast consensus bundles (z, y, s) only when state prediction error exceeds the MOCPO threshold, swarm-wide channel congestion is curtailed, preserving critical wireless spectrum for ground data offloading.

V. ARCHITECTURAL DEPLOYMENT GUIDELINES

Based on the quantitative trade-offs synthesized across the 15 combinations, we formulate the following operational recommendations for aerial edge computing networks:

- 1) **Ultra-Low Latency & High-Frequency Control** ($\Delta t < 10$ ms): Deploy **DYNAMIC** paired with **PPO** or **TACC**. DYNAMIC's sub-0.1 ms execution is the only scheduler fast enough to fit within micro-second hardware telemetry loops.
- 2) **Maximum Throughput & Wide-Area Surveillance:** Deploy **CAR** paired with **LEXICOGRAPHIC** or **MARL**. The global Otsu region partitioning maximizes aggregate ground coverage, achieving up to 35.73% sustained throughput.
- 3) **Bandwidth-Constrained & Energy-Critical Missions:** Deploy **TACC-AAV** paired with **DYNAMIC**. This combination halves wireless broadcast traffic (50.0% reduction) while ensuring peak deadline slack margins (0.392) via dynamic battery headroom regulation.

VI. CONCLUSION

This paper presented a comprehensive empirical investigation into the joint optimization of trajectory planning and task scheduling in multi-UAV FANET-MEC networks. By systematically cross-multiplying three trajectory algorithms (PPO, CAR, TACC-AAV) with five scheduling paradigms (DYNAMIC, MOGS, LEXICOGRAPHIC, BIDDING, MARL) across 100 epochs, we demonstrated that trajectory geometry dictates system capacity and workload equity, while matching maintenance algorithms dictate runtime feasibility. The proposed DYNAMIC repair scheduler achieves near-constant sub-millisecond execution overhead, outperforming deep reinforcement learning and auction baselines by up to two orders of magnitude without sacrificing fairness. Future work will investigate 3D obstacle avoidance in urban canyon topologies and co-training trajectory policies with dynamic repair states.

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